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Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 1 – Réseaux 2

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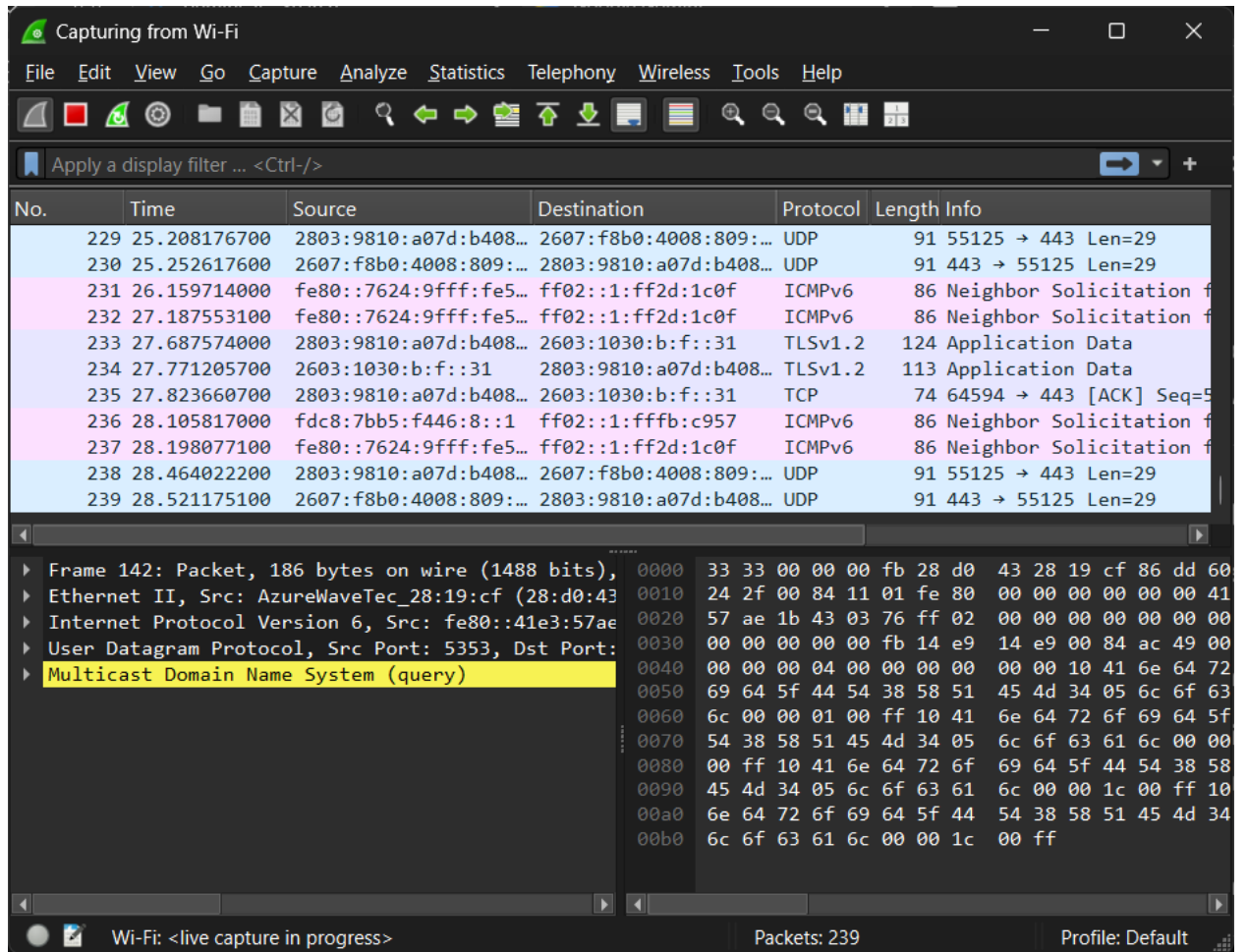
Niveau : L3

Date: 22/03/2026

L'objectif de ce TD est de :

1. Comprendre le rôle et le fonctionnement de Wireshark.
2. Capturer et analyser le trafic réseau sur une interface donnée.
3. Identifier différents protocoles et comprendre leurs interactions.
4. Détecter des problèmes simples de réseau et analyser des communications réseau.
5. Comprendre comment Wireshark peut aider à détecter des intrusions réseau.
6. Identifier des comportements suspects ou anormaux sur un réseau.
7. Analyser différents types d'attaques réseau simples : scans de ports, ping flood, tentatives d'intrusion HTTP

❖ On a l'interface du Wireshark



❖ IP Adress dans Powershell : ipconfig

```
Windows PowerShell
PS C:\Users\Brian ANTOINE> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::6b7a:2a17:553d:8973%20
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : 2803:9810:a07d:b408:bfbf:3be5:3098:9855
    IPv6 Address. . . . . : fdc8:7bb5:f446:8:4bba:a699:f4a9:af55
    Temporary IPv6 Address. . . . . : 2803:9810:a07d:b408:3da5:68d0:4aca:6725
    Temporary IPv6 Address. . . . . : fdc8:7bb5:f446:8:3da5:68d0:4aca:6725
```

```
Windows PowerShell

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::6b7a:2a17:553d:8973%20
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : 2803:9810:a07d:b408:bfbf:3be5:3098:9855
    IPv6 Address. . . . . : fdc8:7bb5:f446:8:4bba:a699:f4a9:af55
    Temporary IPv6 Address. . . . . : 2803:9810:a07d:b408:3da5:68d0:4aca:6725
    Temporary IPv6 Address. . . . . : fdc8:7bb5:f446:8:3da5:68d0:4aca:6725
    Link-Local IPv6 Address . . . . . : fe80::cf4d:9025:a2f2:29f0%7
    IPv4 Address. . . . . : 192.168.1.124
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::7624:9fff:fe56:ea90%7
                                192.168.1.1

PS C:\Users\Brian ANTOINE>
```

❖ Avec mon adresse ip sur wireshark :

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.addr == 192.168.1.124

No.	Time	Source	Destination	Protocol	Length	Info
21	1.348593700	52.113.194.132	192.168.1.124	TCP	54	443 → 56665 [RST, ACK]
119	11.511472600	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
120	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
121	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
122	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
123	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
128	11.556376100	192.168.1.124	142.251.35.238	TCP	54	62813 → 443 [ACK] Seq=4
132	11.584602000	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=8
326	44.209532600	192.168.1.124	40.126.24.84	TCP	54	51035 → 443 [FIN, ACK]
327	44.513369700	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51
334	45.123227900	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51
337	46.234433400	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51

Frame 122: Packet, 54 bytes on wire (432 bits), 5
 Ethernet II, Src: 2a:c8:7b:25:f4:46 (2a:c8:7b:25:0010), Dst: 00:00:00:00:00:00
 Internet Protocol Version 4, Src: 142.251.35.238, Dst: 192.168.1.124
 Transmission Control Protocol, Src Port: 443, Dst Port: 56665

wireshark_Wi-FiZ3QM3.pcapng Packets: 19745 · Displayed: 5152 (26.1%) Profile: Default

❖ Pour dns :

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
4577	258.006423800	192.168.1.124	192.168.1.1	DNS	84	Standard query 0x8160 A
4654	262.123396400	192.168.1.124	192.168.1.1	DNS	84	Standard query 0xf75e A
4657	262.123628800	192.168.1.124	192.168.1.1	DNS	84	Standard query 0x6b27 A
4738	265.146401700	192.168.1.124	192.168.1.1	DNS	84	Standard query 0xf75e A
4740	265.146556400	192.168.1.124	192.168.1.1	DNS	84	Standard query 0x6b27 A
4809	269.149765200	192.168.1.124	192.168.1.1	DNS	84	Standard query 0x6b27 A
4813	269.150476200	192.168.1.124	192.168.1.1	DNS	84	Standard query 0xf75e A
1933	143.667724800	192.168.1.124	192.168.1.1	DNS	85	Standard query 0xda85 A
1934	143.667724700	192.168.1.124	192.168.1.1	DNS	85	Standard query 0x1c72 A
515	82.378240600	fd8c:7bb5:f446:8:3d...	fd8c:7bb5:f446:8::1	DNS	86	Standard query 0x96f1 A
516	82.378808500	fd8c:7bb5:f446:8:3d...	fd8c:7bb5:f446:8::1	DNS	86	Standard query 0x5e6c A

Frame 515: Packet, 86 bytes on wire (688 bits), 8
 Ethernet II, Src: FNLINKTECHNO 64:03:ae (64:82:14:00:03:ae), Dst: 01:00:5e:00:00:01
 Internet Protocol Version 6, Src: fd8c:7bb5:f446:8:3d..., Dst: fd8c:7bb5:f446:8::1
 User Datagram Protocol, Src Port: 62700, Dst Port: 53
 Domain Name System (query)

Domain Name System: Protocol Packets: 20631 · Displayed: 1196 (5.8%) Profile: Default

❖ Faisons ping google.com dans CMD

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Brian ANTOINE>ping google.com

Pinging google.com [2607:f8b0:4008:805::200e] with 32 bytes of data:
Reply from 2607:f8b0:4008:805::200e: time=35ms
Reply from 2607:f8b0:4008:805::200e: time=47ms
Reply from 2607:f8b0:4008:805::200e: time=49ms
Reply from 2607:f8b0:4008:805::200e: time=29ms

Ping statistics for 2607:f8b0:4008:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 29ms, Maximum = 49ms, Average = 40ms

C:\Users\Brian ANTOINE>
```

❖ Pour tcp

Wireshark packet capture showing TCP traffic. The packet list shows several TCP segments, including a retransmission. The packet details pane shows the structure of a TCP segment.

No.	Time	Source	Destination	Protocol	Length	Info
21	1.348593700	52.113.194.132	192.168.1.124	TCP	54	443 → 56665 [RST, ACK]
119	11.511472600	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
120	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
121	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
122	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
123	11.515811300	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=1
128	11.556376100	192.168.1.124	142.251.35.238	TCP	54	62813 → 443 [ACK] Seq=4
132	11.584602000	142.251.35.238	192.168.1.124	TCP	54	443 → 62813 [ACK] Seq=8
326	44.209532600	192.168.1.124	40.126.24.84	TCP	54	51035 → 443 [FIN, ACK]
327	44.513369700	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51
334	45.123227900	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51
337	45.224421400	192.168.1.124	40.126.24.84	TCP	54	[TCP Retransmission] 51

Frame 21: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0, Ethernet II, Src: TIBRO_56:ea:90 (74:24:9f:56:ea:90), Dst: 192.168.1.124, Internet Protocol Version 4, Src: 52.113.194.132, Destination: 192.168.1.124, Transmission Control Protocol, Src Port: 443, Dst Port: 56665

Pour http :

Wireshark packet capture showing HTTP traffic. The packet list shows several GET requests to /connecttest.txt. The packet details pane shows the structure of a packet: Ethernet II, Internet Protocol Version 6, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
2422	150.940307000	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	186	GET /connecttest.txt HT
6748	307.975253600	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	186	GET /connecttest.txt HT
20609	1429.4490735...	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	186	GET /connecttest.txt HT
23759	1731.2085634...	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	186	GET /connecttest.txt HT
6173	286.598101200	192.168.1.124	104.89.170.194	HTTP	208	GET /connecttest.txt HT
6174	286.598141200	192.168.1.124	104.89.170.194	HTTP	208	GET /connecttest.txt HT
6718	307.701228900	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	229	GET /connecttest.txt HT
6721	307.701976700	2803:9810:a07d:b408...	2600:1403:9c00:2e::...	HTTP	229	GET /connecttest.txt HT
6181	286.621374200	104.89.170.194	192.168.1.124	HTTP	241	HTTP/1.1 200 OK (text/
6185	286.626167600	104.89.170.194	192.168.1.124	HTTP	241	HTTP/1.1 200 OK (text/
2426	151.012321500	2600:1403:9c00:2e::...	2803:9810:a07d:b408...	HTTP	261	HTTP/1.1 200 OK (text/

Frame 2422: Packet, 186 bytes on wire (1488 bits) 0000 74 24 9f 56 ea 90 64 82 14 64 03 ae 86 dd 60
Ethernet II, Src: FNLINKTECHNO_64:03:ae (64:82:14 0010 30 c7 00 84 06 40 28 03 98 10 a0 7d b4 08 3d
Internet Protocol Version 6, Src: 2803:9810:a07d: 0020 68 d0 4a ca 67 25 26 00 14 03 9c 00 00 2e 00
Transmission Control Protocol, Src Port: 59586, D 0030 00 00 17 db 03 9a e8 c2 00 50 52 07 cb 32 11
Hypertext Transfer Protocol 0040 09 bd 50 18 00 ff c4 31 00 00 47 45 54 20 2f
0050 6f 6e 6e 65 63 74 74 65 73 74 2e 74 78 74 20
0060 54 54 50 2f 31 2e 31 0d 0a 43 6f 6e 6e 65 63
0070 69 6f 6e 3a 20 43 6c 6f 73 65 0d 0a 55 73 65
0080 2d 41 67 65 6e 74 3a 20 4d 69 63 72 6f 73 6f
0090 74 20 4e 43 53 49 0d 0a 48 6f 73 74 3a 20 69
00a0 76 36 2e 6d 73 66 74 63 6f 6e 6e 65 63 74 74
00b0 73 74 2e 63 6f 6d 0d 0a 0d 0a

Hypertext Transfer Protocol: Protocol Packets: 24137 · Displayed: 18 (0.1%) Profile: Default

❖ ON ouvre Statistics > Protocol hierarchy

Wireshark Protocol Hierarchy Statistics - Wi-Fi

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU's
Frame	100.0	18	100.0	4656	23	0	0	0	18
Ethernet	100.0	18	5.4	252	1	0	0	0	18
Internet Protocol Version 6	66.7	12	10.3	480	2	0	0	0	12
Transmission Control Protocol	66.7	12	5.2	240	1	0	0	0	12
Hypertext Transfer Protocol	66.7	12	37.5	1748	8	6	758	3	12
Line-based text data	33.3	6	2.8	132	0	6	132	0	6
Internet Protocol Version 4	33.3	6	2.6	120	0	0	0	0	6
Transmission Control Protocol	33.3	6	2.6	120	0	0	0	0	6
Hypertext Transfer Protocol	33.3	6	22.5	1049	5	3	548	2	6
Online Certificate Status Protocol	5.6	1	10.1	471	2	1	471	2	1
Line-based text data	11.1	2	0.9	44	0	2	44	0	2

Display filter: http

Close Copy Protocols Help

❖ On ouvre Statistics> conversations>identifier l'IP qui le plus de paquets

Wireshark - Conversations - Wi-Fi

Conversation Settings:

- ☐ Name resolution
- ☐ Absolute start time
- ☐ Display raw data
- ☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

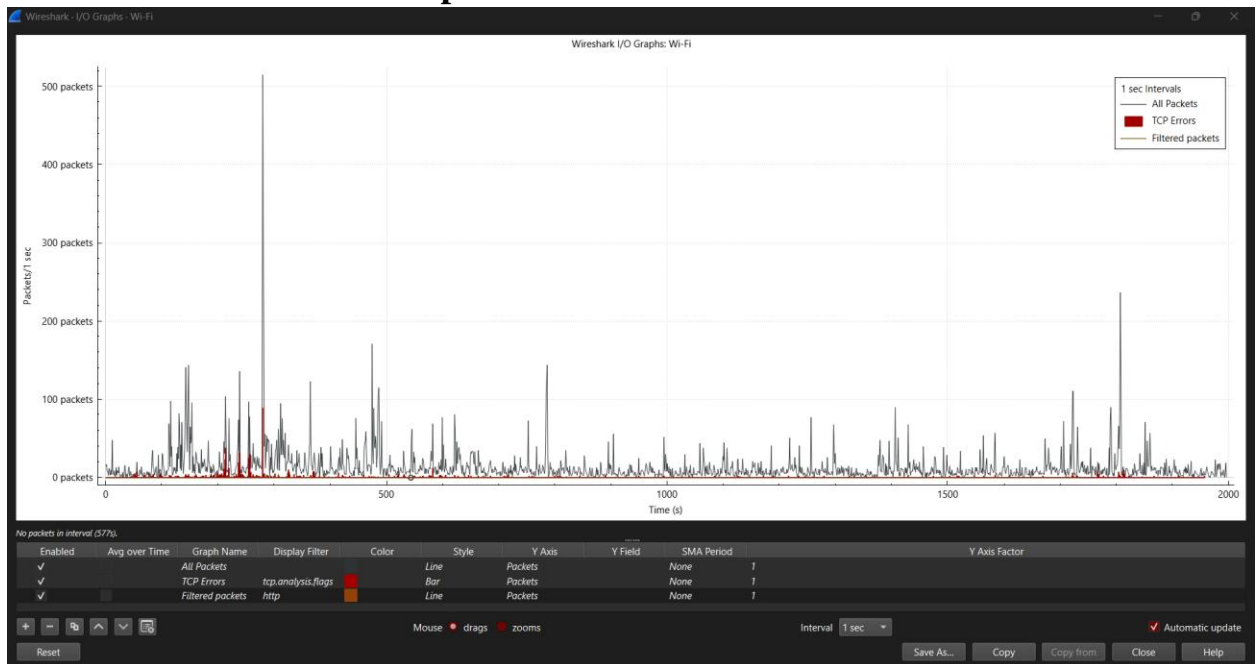
- Bluetooth
- BPv7
- DCCP
- DNP 3.0
- ☒ Ethernet
- FC
- FDDI
- IEEE 802.11
- IEEE 802.15.4
- ILNP

Filter list for specific type

Ethernet - 2	IPv4 - 2	IPv6 - 3	TCP - 9	UDP	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s
2ac2:7b25:1446	64:82:14:64:03:ae		7	2 kB	18	2,246	0.31%	7	2 kB	0	0 bytes	11.515811300	1914.126750	9
74:24:9f:56:ea:90	64:82:14:64:03:ae		11	2 kB	0	10,234	0.11%	2	522 bytes	9	2 kB	0.000000000	1942.857085	2

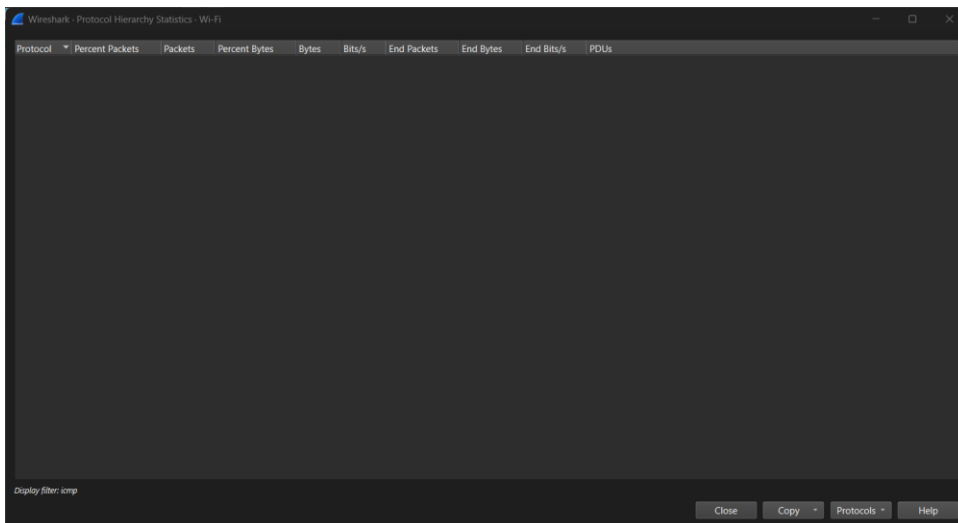
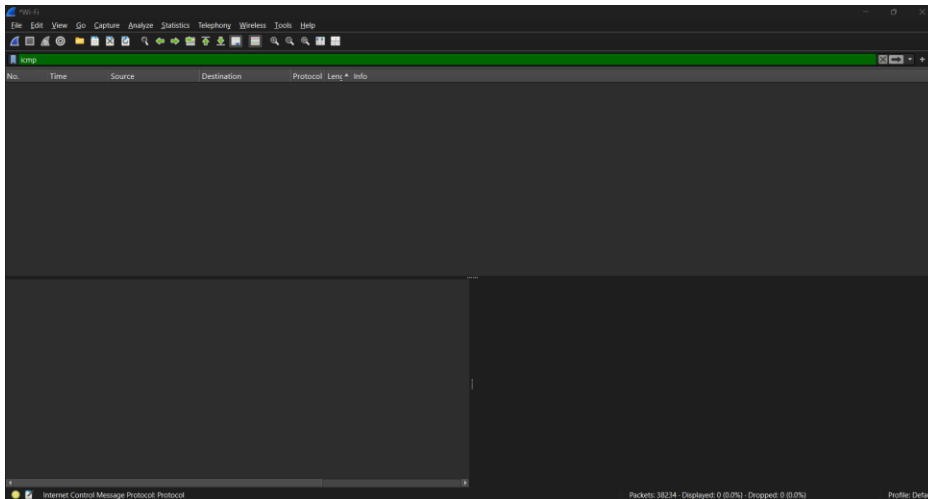
Close Help

❖ Ouvrons statistics>I/O Graphs >>



Inconvénients :

L'objectif de ce TD était de comprendre le rôle de Wireshark dans la capture et l'analyse du trafic réseau. Lors de l'exécution, j'ai pu identifier avec succès les sessions TCP (handshake SYN/ACK) et les requêtes HTTP. Cependant, concernant le protocole ICMP (utilisé pour le ping), aucun paquet n'a été détecté dans la capture malgré l'envoi des commandes. Cette absence s'explique par un filtrage rigoureux du pare-feu de la machine cible ou du réseau, qui bloque les requêtes Echo Request pour prévenir les scans de ports et les intrusions. Ce constat illustre concrètement l'un des points clés du TD : comprendre comment les configurations réseau permettent de détecter ou d'empêcher des comportements suspects. Enfin, l'utilisation des outils statistiques comme le Protocol Hierarchy et les I/O Graphs reste essentielle pour visualiser tout pic de trafic anormal, confirmant ainsi la réussite de l'apprentissage des méthodes d'analyse de sécurité.



Conclusion

Ce TD m'a permis de me familiariser avec **Wireshark** en tant qu'outil d'analyse de trafic en temps réel. J'ai appris à filtrer et à interpréter les protocoles fondamentaux du réseau